

pypresso on a GPU against Quantum ESPRESSO

twelve cases across the physics, one H200 against one CPU core

generated by `performance/plot_gpu_sweep.py` from `gpu-sweep.json`

2026-08-26

Read the baseline before the ratio

This is the comparison GPU.md §2.3 rules out by default, and the reason is not pedantry. The project’s standing metric is single-core pypresso against single-core Quantum ESPRESSO, so that a ratio measures *code*. A GPU number against a serial CPU number measures code *and* hardware at once and cannot be decomposed afterwards. It is reported here because it was asked for, with the baseline named in every row.

Two further things bound how far these numbers travel:

- **One core is the softest baseline available.** For a single k -point `pw.x` has only plane-wave parallelism and saturates early — measured elsewhere in `PERFORMANCE.md`, 16 to 32 cores buys 4%, and against 32 cores the same silicon comparison falls to about 1x.
- **The iteration counts differ, in both directions.** pypresso takes fewer iterations than QE on silicon and five times as many on `ni10-ldau`. That is mixing and starting guess, not hardware, and it would show identically on a CPU. *The per-iteration column is the code comparison; the total column is time-to-answer.*

GPU	NVIDIA H200 (141 GB)
CPU baseline	AMD EPYC / Intel Xeon Cascade Lake, 1 core
reference code	Quantum ESPRESSO 7.2 (serial, 1 MPI task)
pypresso	commit <code>c5dc7d4</code> , JAX 0.11.1 + <code>jax-cuda12-plugin</code> 0.11.1
settings	pypresso: <code>k_batch=all</code> , <code>band_batch=all</code> , <code>conv_thr = 1e-10</code>
date	2026-08-26

What it costs

The speedup is not one number, and what it tracks is work per k -point. It runs from 2.1x on `al10-metal` — a cheap cell where a single core is already efficient and the GPU is mostly idle — to 115.5x on `bi20-soc`, where QE spends 105s *per iteration* on two-component spinors with a fully-relativistic ultrasoft dataset and the GPU spends 0.91s. The ordering in Figure 1 is the physics: norm-conserving and metallic cells at the bottom, ultrasoft, PAW and magnetism in the middle, spin-orbit at the top.

Whether it is right

Every case that does not involve spin-orbit agrees with Quantum ESPRESSO to 4.6e-09 Ry or better, and those figures reproduce the *CPU* agreements already recorded in `PLAN.md` — 1.9e-9 on `al10-metal`, 4.3e-9 on `h10-chain-ldsa`, 4.4e-9 on `h10-chain-noncolin`. **The GPU reproduces pypresso’s behaviour, including its known imperfections, rather than having a numerical character of its own.**

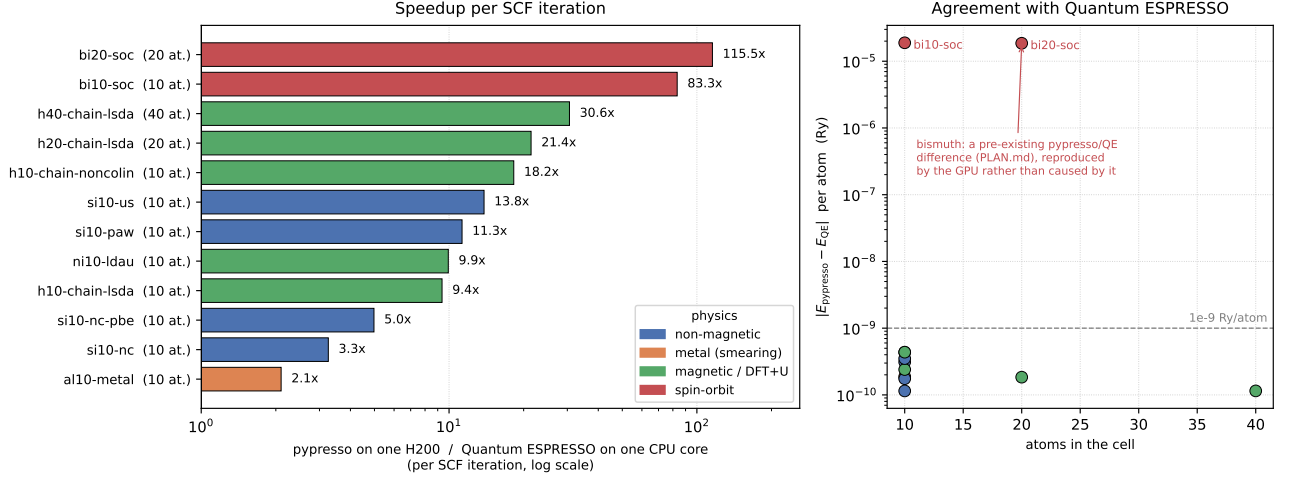


Figure 1: **Left:** speedup per SCF iteration, sorted, on a log axis because it spans two decades. **Right:** the agreement the timings rest on. The colour is the physics in both panels.

case	physics	at.	k	QE, 1 core		pypresso, H200		speedup	
				it.	ms/it	it.	ms/it	per it.	total
si10-nc	norm-conserving	10	7	14	98	8	30	3.3x	5.7x
si10-nc-pbe	GGA (PBE)	10	7	16	221	8	44	5.0x	10.0x
si10-paw	PAW	10	7	14	644	8	57	11.3x	19.7x
si10-us	ultrasoft	10	7	15	595	8	43	13.8x	26.0x
al10-metal	metal, smearing	10	10	47	266	10	127	2.1x	9.9x
h10-chain-lda	collinear magnetic	10	2	12	1219	17	130	9.4x	6.6x
h10-chain-noncolin	noncollinear	10	2	13	2942	17	161	18.2x	14.0x
ni10-lda	DFT+U, magnetic	10	10	29	3191	151	321	9.9x	1.9x
h20-chain-lda	collinear magnetic	20	1	15	2991	33	140	21.4x	9.7x
h40-chain-lda	collinear magnetic	40	1	22	12669	104	414	30.6x	6.5x
bi10-soc	spin-orbit	10	1	14	21686	17	260	83.3x	68.6x
bi20-soc	spin-orbit	20	1	20	105350	20	912	115.5x	115.5x

Table 1: Every case converged to conv_thr = 1e-10. QE’s times are its own timing report (electrons), not a stopwatch around the process; pypresso’s are the warm SCF, with compilation reported separately and never amortised into the loop.

case	at.	QE total energy (Ry)	pypresso (Ry)	ΔE (Ry)	peak GPU (GB)
si10-nc	10	-78.99748976	-78.99748976	-3.1e-09	0.18
si10-nc-pbe	10	-78.65632709	-78.65632709	-1.7e-09	0.39
si10-paw	10	-446.39712348	-446.39712348	-3.5e-09	0.66
si10-us	10	-113.78984345	-113.78984345	-1.1e-09	0.65
al10-metal	10	-41.85395892	-41.85395892	-1.9e-09	0.40
h10-chain-lda	10	-9.56782281	-9.56782281	-4.3e-09	0.55
h10-chain-noncolin	10	-9.56782281	-9.56782281	-4.4e-09	1.37
ni10-lda	10	-856.58261082	-856.58261082	+2.4e-09	1.60
h20-chain-lda	20	-19.13564562	-19.13564562	+3.7e-09	1.10
h40-chain-lda	40	-38.27138667	-38.27138667	+4.6e-09	4.06
bi10-soc	10	-1477.73706506	-1477.73725394	-1.9e-04 [†]	16.92
bi20-soc	20	-2955.16301232	-2955.16338417	-3.7e-04 [†]	34.70

Table 2: [†] The bismuth rows are a *pre-existing* pypresso/QE difference, recorded in PLAN.md as “the one that does not close” at 1.9e-4 Ry, not something the GPU introduces — and QE 7.5 and 7.2 agree with each other to 1e-8 on that case, so it is not a version effect either.

Two things worth carrying

Memory is the constraint that bites, and spin-orbit is where it bites. `bi20-soc` peaked at 34.7 GB: two-component spinors on a fully-relativistic ultrasoft dataset. It fits a 141 GB H200 with room to spare and *would not fit a 32 GB V100*, which is the sort of fact that decides whether a calculation exists rather than how fast it is.

An iteration cap is not a convergence failure. The first run of this sweep reported `ni10-ldau` and `h40-chain-lda` as not converging on the GPU. They were hitting `run_scf`'s default of 100 iterations, where the regression suite has always used 200 for exactly those cases; `ni10-ldau` reproduces the same non-convergence on a CPU at 100, which is what settled it. Both converge here, at 151 and 104 iterations.